Benefits and challenges of consortium blockchains

Consortium blockchains offer several benefits but face some unique challenges.

Benefits

- *Security and transparency:* Blockchain is a secure and transparent way to store data. The data is encrypted and distributed across the network, making it difficult to hack or tamper with. The distributed ledger also makes it easy to track the history of transactions, which can help to prevent fraud.
- *Collaboration:* Consortium blockchains can help to facilitate collaboration between organisations. By sharing data and processes on a shared platform, organisations can work together more efficiently and effectively.
- *Cost savings:* Consortium blockchains can help to reduce costs. By sharing resources and infrastructure, organisations can save money on development, maintenance, and operation costs.
- *Scalability:* These blockchains can be scaled to handle a large number of transactions. This makes them a good fit for applications that require high throughput, such as supply chain management and trade finance.

Challenges

- *Governance:* Consortium blockchains require a well-defined governance framework to ensure that all participants agree on the rules and procedures for the network. This can be a challenge, especially if the consortium includes organisations with different interests.
- *Complexity:* These blockchains can be complex to implement and manage. This is because they require a high level of technical expertise and coordination between the participants.
- *Regulation:* The regulatory landscape for blockchains is still evolving. This can make it difficult for organisations to comply with all applicable regulations.
- *Lack of standards:* There are no universally accepted standards for consortium blockchains. This can make it difficult for organisations to interoperate with each other.

Overall, consortium blockchains offer a number of benefits, but they also come with some challenges. The decision of whether or not to use a consortium blockchain should be made on a case-by-case basis, taking into account the specific needs of the organisation.

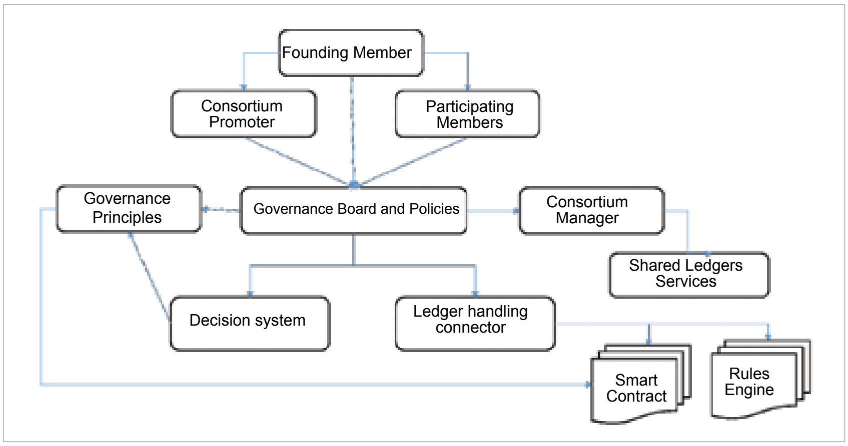

Figure 2: Consortium blockchain governance

Consortium blockchain governance

The functional architecture of any blockchain platform, which helps to define the governance model, is based on various consensus algorithms used for platform execution. These are listed here.

Proof of work (PoW)*:* This is the original consensus algorithm from Satoshi Nakamoto's first research on blockchain, and is used as a mechanism to confirm the transaction has been completed. If confirmed, a new block gets added to the chain of networks. This helps prevent distributed denial of service (DDoS) network attacks.

***Proof of stake* (PoS):** PoS involves a lot of system resources like GPUs. Hence, it is an advanced mechanism suitable for sophisticated blockchain networks. PoS addresses the challenge of high resource usage by using an algorithm based on distributed tokens and static coin supply.

Delegated proof of stake (DPoS): This is an extended version of PoS. PoS is like a lottery mechanism based on the coin stake of the users, while DPoS influences all stakeholders in the network equally.

Consortium blockchains have become popular in the last few years. They are widely preferred as a solution to replace legacy distributed workflow systems like simplified document management systems or semi-manual compliance systems.

Typically, in a consortium blockchain, we have various key entities (Figure 2) as explained below.

- The **founding member** develops the governance model and hierarchy of groups to be used in the blockchain platform.
- The **consortium promoter** addresses the administrative activity for implementing the blockchain platform of choice — R3, Hyperledger, B3i, Bankchain, etc — depending on the industrial solution and use case being addressed.
- The **participating members** are the actual users of the blockchain platform. They handle the transactions and prepare the interconnection between nodes in the blockchain platform.